then started having access to bodies, would do
a lot more to learn about the bodies
- Also got rid of God / would do so

9 Materialism: The Enlightened Machine

Annoying

Humans are really mechanical beings

The Vexing Alternative - don't need soul

In the two preceding chapters, we have reviewed the philosophical psychologies of the English and Continental empiricists and rationalists, confining ourselves to the seventeenth and eighteenth centuries, when these become sufficiently definite to serve as alternatives. In Chapter 3, the very considerable similarities between Platonist and Aristotelian systems were noted. In subsequent chapters, equivalent similarities were found between philosophers identified with one or another school. In the Renaissance, and really not until the Renaissance, did we find signs of a clear break, the break between spiritualism and naturalism. But even in the Renaissance there were only signs, not dramatic ruptures. The great separation took place in the irresolvable conflicts between Hume's *Treatise* and Kant's *Critique*. Henceforth, empiricism and rationalism would hold out possibilities for radical departures and divisions. Unlike the differences that divided Greek atomists and Platonists, the modern antagonisms would surface as completed systems of thought, rich in implications and recommendations for social organization, law, morality, economics, and religion. By comparison, the disputes between orthodox Christians and Ockhamists in the thirteenth century or the Florentine Aristotelians and Platonists in the fifteenth would be nearly negligible.

As great as the break between rationalism and empiricism was, however, there remained ground common enough for a Kant even to attempt to answer a Hume. The philosophical disputes of the seventeenth and eighteenth centuries, to the extent that they were between rationalists and empiricists, were disputes about the same kinds of problems, and they were disputes that often began from identical starting points. Hume and Kant were both moved to explain the source of human knowledge, the nature of morality, and the character of society. Broadly, they were both engaged in the "science of *mental* life."

Neither of them subscribed to the vexing and alternative: law, morality, reason, and feeling were, in the last analysis, mere expressions of matter in motion. In Chapter 7, we noted that Locke declined to address the question of the human material nature. Berkeley openly rejected materialism and defiantly so. Hume, while tilting with the possibility, finally turned the matter over to the "anatomists."[1] Kant, too, recognized that a biological interpretation of his philosophy might be advanced, that his categories might be treated as rising from a neurological process, and he spurned the suggestion, noting that such a skeptical materialism would remove the element of necessity from the pure concepts of the understanding.[2] In short, neither Kant nor Hume nor any of the earlier figures in the two philosophical movements conceived of psychology as indistinguishable from physics. For all of them, it was and must remain a science of *mental* life and not merely science. However, in the same two centuries, a third movement was born and flourished: a movement toward physics and away from logic, a movement that came to reject the very terms of the rationalist-empiricist controversy, a movement that had more to do with the founding of psychology as a scientific discipline than did all the rationalists and empiricists combined. This was materialism, whose seventeenth- and eighteenth-century character is the subject of the present chapter.

The Metaphor of the Machine

Every age of philosophical energy is animated by notions and events of a nonphilosophical complexion. Philosophers seek to understand and explain the facts of the world and they must take these facts as they find them. They are to be found, of course, outside philosophy: in the cosmos, in the world of matter, in the human mind, in the affairs of state.

Scholarship is a human enterprise and, no matter how narrow and technical it may become, or how ponderous or cultist its problems and methods may be, it seldom escapes the habits of the human mind. One of the most persistent of these habits is that which forces the mind to metaphor and simile when it seeks to comprehend an elusive phenomenon. And, among the many elusive phenomena, none is endowed with greater craft and agility than the mind itself. Thus, in their tireless attempt to comprehend the mind, philosophy and, later, psychology have taken recourse to explanations of the sort, "it is like . . ." or "it is as if . . ." or "it is no more than. . . ." Over the centuries, different metaphors and similes have gained and lost popularity. The pre-Socratics, with their special interest in hydraulics and hydrostatics and with their simplified four-element physics, were given to believe that psychological phenomena were to be understood in terms of the unique combination of earth, air, fire, and water. Platonism, which never fully extricated itself from its roots in the mystery religion of the Pythagoreans, focused on spiritual metaphors in an attempt to define that ineffable quality of mental life. Aristotle, surveying the unchallengeable truths of the syllogism, and noting the contingent nature of all nonlogical realities, advanced a dualistic psychology according to which some functions of the mind were "like" biological processes and others "like" eternal, logical verities. Accordingly, while the nutritive, sensitive, and locomotor faculties perish with the flesh, being of the flesh, the intellect survives "like" the truth of logic. Imbued w/ Spirit = God

From the pre-Socratic period until the Age of Faith, the metaphor was nature. All the divisions and antagonisms of competing philosophies are to be understood as different views of the nature of nature and as differences of opinion regarding the ultimate nature of nature: Is it material only or spiritual as well? Is it eternal or was there a beginning? Is it particulate and statistical or an essential and invariant form? Is it as perception records it or is experience mere illusion? The failure of philosophy to answer these questions and the failure of the very civilizations that raised such questions to survive were two of the chief causes of the rapid success of the Christian alternative. In the Western world, from the Patristic period until the seventeenth century, revelation was the ultimate authority. God was the reality and nature was the metaphor. This is not to say that everyone shared this outlook. Indeed, Chapters 5 and 6 were devoted principally to those who sensed that the question was hardly settled. But the world is peopled by more than philosophers, and for all but the handful of seekers after truth, the Truth was revealed in the gospels and in the life of Christ. Those eager to understand either at another level could study the great Thomistic synthesis—the synthesis of Aristotle and Christian belief, of reason and faith. - now Atheistic Alt

We have pointed out, perhaps too often, that the Renaissance did not alter the essential temper of the Western mind as it grappled with the abiding questions. The major debates from 1350 to 1600 were between those who found God in "natural magic" and those who found Him in "spirit." Ficino's Academy sought to restore Platonism to Christianity, not to challenge scripture. Pomponazzi's Aristotelian school had no quibble with the Bible, nor were the heresies of Gianfrancesco Pico of a religious sort. And this speaks only to the affairs of academics. In the world at large, there were more important matters: war, pestilence, Reformation, starvation. Nature, as the metaphor, and God, as the reality, remained in their historic places.

The most significant scientific event of the Renaissance, Copernicus' theory of the earth's revolution, was judged by its author as no more than a footnote to God's great design. elliptical orbits The same may be said of Kepler's assessment of his theory and Newton's of his. It is equally true of Galileo and of Locke and Descartes, of Leibniz and perhaps even Spinoza. The greatest scholars in philosophy and science, even through much of the nineteenth century, labored under the light of Christian faith. Hume, of course, is the tantalizing exception, an exception even among seventeenth-century skeptics, of which there were many. The skeptics trained their doubts on man, not on God; on Aristotle, not on the Nazarene. In fact, several of the noteworthy skeptics were priests.

In light of the foregoing, it is instructive to examine a remarkable feature of contemporary psychology: no major spokesperson for the discipline, no figure identified as one responsible for its methods and concerns, none who has provided a theory of consequence to contemporary endeavors, has argued that the religious dimension of life is necessary for an understanding of human psychology. Stated another way, we recognize that in the fifteen centuries beginning in A.D. 200, there is no record of a serious psychological work devoid of religious allusion and that, since 1930, there has not been a major psychological work expressing a need for spiritual terms in an attempt to comprehend the psychological dimensions of man. This striking shift in perspective cannot be explained as a product of rationalist-empiricist tensions—witness Locke and Leibniz. Nor is it to be understood in terms of some factual discovery in the natural sciences. Moreover, it cannot even be understood as a gradual shift in perspective because, in the historical measure of time, it has been sudden. We are not able to establish all the causes, of course, but it is fair to say that we can discern its origin. The origin of this dramatic transformation in perspective, this historically unprecedented abandonment of an older and pervasive vision, is the seduction of a different metaphor: in this case, the metaphor of the machine.

The cautious reader will be ready to complain that the physicalistic outlook is hardly recent; that Zeno and Epicurus were beholden to it; that Ptolemy rendered it cosmic in scope; and that the Scholastics were as wed to the machine-like precision of celestial motion as were the pre-Socratics. While this is a correct reading of history, it fails to distinguish between the mere metaphor and the seductive metaphor: between the metaphor that confirms and conforms to the balance of one's beliefs and the metaphor that creates a new belief, and between the metaphor invented to represent reality and that which is so compelling as to be accepted as reality. We must ask, then, what there was in the mechanistic philosophy of seventeenth- and eighteenth-century thought that set the stage for the radical divorce now existing between psychology and its intellectual ancestry. What new feature was added or recognized? Who fashioned the wedge? We can only begin to answer this by discussing, all too briefly, a set of sociopolitical conditions and, coincident with those conditions, the extraordinary influence of Galileo on his contemporaries and immediate successors.

The political climates of England and France in the seventeenth and eighteenth centuries were touched upon in the preceding two chapters. The issues raised by the Reformation and the responses to these during the counter-Reformation remained hopelessly and perilously unsettled. Religious persecu-

(the church loses its power).

tions would continue in both countries until the nineteenth century, and religious toleration—always difficult to gauge—would become a political maxim only at the end of that century. Added to the death struggle of Protestant and Catholic, Protestant and Protestant, and Catholic and Catholic were the endless conflicts within the fraternity of European nations. The Roman Church, which had tied its fortunes to Aristotelianism, as understood by the Scholastics, found it necessary to take hardened stands on philosophical matters at a time when intellectual freedom was coming to rank as high as life itself in Paris. Descartes' *Method,* based on doubt and skepticism as points of departure, came closely to heresy. The writings of Montaigne, filled with Erasmian charm and insolent wit, were proscribed, making them even more popular than would otherwise have been the case. The murder of Ramus, the famous anti-Aristotelian at the University of Paris, gave the liberal-leaning philosophers of the early seventeenth century still one more reason to resist the authority of Aristotelianism. But unlike all their predecessors, they had an irrefutable, scientific illustration of the failure of the philosopher's *Metaphysics:* Galileo's laws of accelerated motion and Newton's general laws of motion.

Among the Thomistic proofs for the existence of God, the phenomenon of motion was central. Relying on the Aristotelian argument for a prime cause, the scholastic proof began with the contention that that which is in motion will seek a resting state unless impelled by a force external to itself: that is, eternal motion is impossible, and, therefore, heavenly dynamics can be understood only in terms of a Prime Mover who constantly supplies the needed power. The planets move by the will of God, a proof providing a ready metaphor for human action as the result of human will. If the action of the planets required no recourse to the will of God, the actions of men might not either. More will be said on this later. What Newton demonstrated was that bodies set in motion would continue to move, linearly and eternally, unless acted upon by a force opposed to their motion. Quite simply, once the bodies were set in motion, God's will had no further work to do for the motion to continue forever. God aside—for it was not God to whom Newton addressed his remarks—*Aristotle was wrong*. He was also wrong, and Galileo proved it, in contending that the speed with which an object fell to the ground was proportional to its mass (weight). He was also wrong about the immobility of the earth and about the motion of the sun. (Recall, however, that Aristotle was far from dogmatic on this point.)

Had Galileo's contribution been limited to the unearthing of mere factual errors in the corpus of Aristotelian science, his influence on his own age and ours would have been negligible. The health of Christian belief never depended on the notions of a wise but pagan Greek about the speed with which rocks and feathers fall to the earth. It depended even less on the number of moons orbiting Saturn since, no matter how many there are, they all came from the same shop.

the Path to the truth (how ask questions)

However, orthodox belief did depend on the manner in which we go about *discovering* the truth, and it was this that Galileo threatened.

In his *Discourse on the Two New Sciences,*[3] Galileo virtually invented the science of mechanics, passed it on in a form that has hardly been changed, and tied each of its theorems and propositions to an *experimental* demonstration. Throughout his work, Aristotle falls on hard times, even when credited with a truth. For example: Aristotle had some idea of the principle of the lever—force bears to resistance a relationship determined by the reciprocal of the distances separating the fulcrum from the force and the resistance—derived and demonstrated by Archimedes. Galileo acknowledged Aristotle's observation on the law of the lever, but with a qualification, as usual:

> Yes, I am willing to concede him priority in point of time; but as regards rigor of demonstration the first place must be given to Archimedes.[4]

The most open attack on Aristotelianism appeared in Galileo's *Dialogue on the Two Great Systems of the World*[5] (1632). The year following its appearance found its author before the Inquisitors and called upon to renounce the heresy of Copernicanism. Copernicus was dead now for ninety years, and Galileo was not the only philosopher-scientist to subscribe to his theory. We may take it, then, that it was not merely the Copernicanism of the *Two Great Systems of the World* that troubled the Inquisitors but the ridicule heaped on the Aristotelian position in the dialogue.

Kepler (1571–1630) had advanced three laws of planetary motion remarkable in their simplicity and predictive power, and these laws assumed the earth's elliptical movement around the sun. Galileo (1564–1642) accepted the Copernican-Keplerian theory and added to it a completed body of terrestrial mechanics. William Harvey (1578–1657), who had studied with Galileo and the other great Paduans from 1598 to 1602, published his famous essay on the circulation of blood in 1628. It is to be noted that Harvey was Francis Bacon's physician—noted as a curio of history—but Harvey is to be seen as a disciple of Galileo's scientific method: one begins with a careful measurement of nature as it is found; one advances the simplest possible hypothesis sufficient to account for the measures obtained; one derives from the confirmed hypothesis those corollaries logically implied by the hypothesis; one proceeds to test the corollaries experimentally. The opinions of others, no matter how grand their genius may be, count for nothing. Truth is to be found after a test, not at the end of a dispute. The inscription adopted by the Royal Society, formed in 1662, is a testimony to Galileo's influence: *Nullius in verba.* ("*On the words of no man.*")

By far the greatest contribution to science in the seventeenth century, if not in any century, was Isaac Newton's *Principia.* Newton (1642–1727) was the saint of seventeenth-century British intellectuals, a man whose influence spread far beyond the boundaries of science. He combined the theories of Kepler and Galileo in a triumphant synthesis, the universal law of gravitation. No longer was it necessary to require, as the Aristotelians did, that a different physics be applied to the heavens and the earth. Galileo had concluded the dialogue of the *Second Day* portentously:

> We do not encounter . . . difficulty, however, if we suppose the earth to move, a body so mall, so inconsiderable in comparison with the whole universe that it could have no effect at all upon this.[6]

Newton, framing no hypotheses except those demanded by the direct observation, placed this inconsiderable ball within the realm of many other balls and kept them all suspended by a law indifferent to human vanity, belief, or hope. In Book III of his *Principia,* he offers the "rules of reasoning" in philosophy:

> Rule I:
> We are to admit no more causes of natural things than such as are both true and sufficient to explain their appearances.
>
> - minimum explanation possible — what you see is what you get
>
> Rule II:
> Therefore to the same natural effects we must, as far as possible, assign the same causes . . .
>
> Rule IV:
>
> null hypothesis — if not disconfirmed, hold as true until proven theory or better way comes along
>
> In experimental philosophy we are to look upon propositions inferred by general induction from phenomena as accurately or very nearly true, notwithstanding any contrary hypotheses that may be imagined, till such a time as other phenomena occur by which they may either be made more accurate or liable to exceptions.[7]

Between 1609 and 1686—that is, in a span of a lifetime—Kepler had published his laws of planetary motion (1609), Galileo had described sun spots, the moon's rough surface, the "stars" of Jupiter (1609), and had invented *Mechanics* (1632), Harvey had published his *Excertatio* (1628), Descartes his *Discourse on Method* (1637), and Robert Hooke the *Micrographia* (1665), which laid the ground for microscopic anatomy. If we wonder what the implications of this creative storm seemed to be to the enlightened bystander, we need only recall Voltaire's remark in *The Ignorant Philosopher:*

> [I]t would be very singular that all nature, all the planets, should obey eternal laws, and that there should be a little animal, five feet high, who, in contempt of these laws, could act as he pleased, solely according to his caprice.[8]

(free will)

The metaphor was becoming reality.

A look at the title

From Dualism to Monism

Descartes' contribution to the scholarship of his day produced results he could not have anticipated. Three years before he died, he received a pamphlet published anonymously in Belgium by a former disciple, Regius. The pamphlet was in a form suitable for nailing on the door of a church and it was written in a way that mimicked Descartes' style. It was a polemic on the nature of the mind and it recommended a radical materialism to those who might inquire into the subject. It offered twenty-one propositions and closed with one of Descartes' own aphorisms: "No men more easily attain a great reputation for piety than the superstitious and the hypocrites."[9] The propositions themselves were uncompromising. The mind is no more than that which permits thought to human beings; logic does not require any distinction between mind and matter; all notions enter the mind through observation and tradition; no idea can be said to be innate; perception is a process of the brain.[10]

Regius arrived at this view through his own inclinations and, probably, from a hasty reading of *Les Passions de l'Âme,* which was completed in 1646. It was not published until 1649 and Descartes regretted its publication. Regius either read the work in an early draft or decided, from conversations with Descartes about the work, that it supported the radical materialist position. Several of the Articles in the book can be so construed. The origins of the animal spirits are traced to the brain such that the turbulence produced by stimulation of the senses becomes reflected in the actions of the muscles. In Article XVI, Descartes notes specifically that stimulation can produce orderly action without any intervention by the soul and in animals that lack souls.[11] Only thought and passion are in need of a soul for their existence. Perception, appetites, and even dreams and daydreams are quite possible as a result of bodily actions.

Descartes quickly recorded his disagreement with Regius's manifesto, rejecting its propositions point by point. It was Regius's twelfth Article that had challenged Descartes on the question of innate ideas, and Descartes' reply is instructive:

> I never wrote or concluded that the mind required innate ideas which were in some sort different from its faculty of thinking; but when I observed the existence in me of certain thoughts which proceeded, not from extraneous objects nor from the determination of my will, but solely from the faculty of thinking which is within me, then . . . I termed (these) "innate."[12]

Descartes goes on in his rebuttal to observe, yet another time, that there is nothing in an object that can be said to *contain* the idea we have of it. Thus, all ideas are innate if only in the sense that they have a quality not derivable from mere extension, which, after all, is the very substance of object:

Monism / body + mind are same substance — brain

> [N]o ideas of things, in the shape in which we envisage them by thought, are presented to us by the senses. So much so that in our ideas there is nothing which was not innate in the mind. . . . Nothing reaches our mind from external objects through the organs of sense beyond certain corporeal movements.[13]

Unlike his arguments in *Les Passions de l'Âme,* which were often of a theological complexion, Descartes, in his rejoinder to Regius, presents the argument of the philosophical dualist: there is nothing in the physical stimulus resembling the mental image, nothing we can say in describing the stimulus that will permit us to deduce its psychological consequences. This argument, as we shall see in subsequent chapters, has lost none of its persuasiveness in contemporary philosophical discourse on the mind-body problem.

The pamphlet printed in Utrecht was a relatively minor irritation. In comparison, the challenges to Cartesianism presented by Pierre Gassendi (1592–1655) were titanic and were the very foundations of that anti-Cartesianism that flowered in the eighteenth century. In his own time, Gassendi was ranked among the greatest philosophers of his age. His following was large and enthusiastic, his writings were influential, and his command of science and mathematics was expert, or at least seemingly so. He established his libertine credentials early, authoring a text of *Paradoxes against the Aristotelians* in 1624. More significantly, he became the center of a revival of interest in Epicurus and later Roman Epicureans. The thrust of this revived Epicureanism was to install observational science in the place of (Cartesian) deductive, "axiomatic" science, to accept nature (including human nature) as matter, and to oppose the authority of Aristotle at every turn. While not a skeptic, Gassendi found it easier to adopt the skeptical posture when the only alternative was dogma. The orientations were marshaled for a critical attack on Cartesianism, one lasting six years and taking the form of published rebuttals answered, in print, by Descartes. We will not examine any part of the dispute other than that addressed to Descartes' dualism. It is worth noting, however, that Gassendi's critique was all-embracing and that, as Professor Craig Brush has written, "Modern philosophy can add little or nothing significant to the objections made by Gassendi."[14]

It is in the Second and Sixth Meditations that Descartes presents his *psychophysical* dualism and it is in the rebuttals to these Meditations that Gassendi offers the monist alternative. Against the principal arguments for dualism, Gassendi registers the Epicurean complaints: (1) Why deny bodies the power to move themselves without the aid of a soul? Does not water flow and do not animals (judged to be without souls) walk?[15] (2) Since it is obvious that whatever *acts* is, why all the "beating around the bush" to establish that you (Descartes) are?[16] (3) What does it mean to say that you are "only a thinking

ds?
Why can't we?

thing"? Why exclude all other possibilities, such as you are a wind or a gas or a body? Even if it be granted that you only *know* you are a thinking thing, it does not necessarily follow that you are not other things as well.[17] (4) How can the mind reason without a brain, since even you (Descartes) have required that the brain organize and unite perceptions and actions?[18] (5) Even though the senses sometimes deceive, they often do not, and we usually have the means of determining whether a given perception is valid or of questionable validity.[19] (6) If the mind without extension, it can have no idea of extended things. The mind can be furnished by experience only if equipped, by its material nature, to respond to that which is physical. It may be said that you are composed of two bodies, one coarse and one subtle, and that only the former is immediately apparent. But it makes no sense to say that *you* or your mind is unextended.[20]

Even more than the specific objections to Descartes' dualism, Gassendi's writings reflect the impatient tone that every modernist displays toward a departing age. Gassendi was the most illustrious of the circle of visionaries that was brought and held together by Father Marsenne in Paris. They were all imbued with the spirit of Galileo, their Paduan saint, whose experimental and theoretical science constituted the final and invincible challenge to authority. Gassendi's works were known to and admired by Locke, whose empirical philosophy borrowed the spirit, if not the letter, of Gassendi's scholarship. Newton, too, acknowledged Gassendi's priority in advancing the law of inertia. While contemporary thought is not directly influenced by any of Gassendi's essays or discoveries, his position among his own contemporaries was one of considerable influence. Descartes' ambition to rest biology on a Keplerian foundation, to establish a mathematics of the life sciences while sparing the soul from materialism, was the ambition of the Gassendists but without the spiritual restrictions. Descartes' desire to combine physics and mathematics into an irrefutable body of knowledge was also an integral feature of the Gassendist program but with the rationalist element replaced by the experimentalism of Galileo. The Gassendists were the first natural monists of the modern era. When we refer to the origins of this perspective as "French Materialism," we are simultaneously acknowledging the role of Pierre Gassendi in the history of scientific psychology. He did less research than he inspired, but his inspiration was nearly without parallel.

☆ The first philosopher that was a materialist

☆ Thomas Hobbes (1588–1679) and the Social Machine

The discussion of Hobbes could have been placed in either of the preceding two chapters as easily as in the present one. Texts devoted to the history of philosophy routinely list him as an early empiricist principally on the basis of such positions as that appearing in Part I of *Leviathan:* With the exception of prudence, which is grounded in experience,

Govt— max pleasure – min pain for majority.
(need Leviathan to make ppl good)

> there is no other act of man's mind, that I can remember, naturally planted in him, so, as to need no other thing, to the exercise of it, but to be born a man, and live with the use of his five Senses.[21]

But, two chapters later, Hobbes goes on to distinguish between sense and reason, noting that the latter is not "gotten by Experience only," and that science, which is a knowledge of consequences, is something more than the mere facts of sense and memory.[22] Thus, while an epistemological empiricist, Hobbes is comfortable with methodological rationalism. In fact, one of the important sources of his scientific thinking was Galileo, whose hypothetico-deductive method provided possibilities utterly lacking in the methodological empiricism of Francis Bacon.

Leviathan appeared in 1651, when its author was already past sixty. Hobbes' earliest writings and training were in classics. His translations of Thucydides were authoritative. For five years (1621–1626) he served as a student-secretary to Francis Bacon and, later, for a time, as a tutor to Charles II. He was involved occasionally in the intellectual affairs of Father Marsenne's "libertines," and it was through them that his visit to Florence and with Galileo was arranged (1634–1637). Although late in appearing, Hobbes' influence on his contemporaries was far from negligible notwithstanding the fact that the list of luminaries active during his long life included Descartes, Locke, Newton, Gassendi, Milton, and Galileo.

Leviathan is a long and uneven work, consuming seven hundred pages of modern print and ranging topically from nutrition to demonology. Our present concerns warrant attention only to those parts of it which advance and defend the materialistic perspective on man and society. The work, as a whole, is suffused with this perspective, but it is in Part One (*Of Man*) that our present interests are most repaid. Before turning to it, however, we might note the general aim of *Leviathan*.

The work appeared two years after the conclusion of the civil war, but before Cromwell had been named Protector (1653) and while English intellectuals were gravely weighing alternative possibilities for government. Hobbes had exiled himself in France, where for a time (1646–1647) he tutored another exile, the future Charles II. The execution of Charles I (1649) was followed by a mixture of lingering anti-royalism and the guilt of regicide. We might say, then, that *Leviathan* was greeted by mixed reviews. It argued for absolute monarchy, but it based the argument not on a king's "divine rights." The restoration of the monarchy placed Charles II on the throne and so, at last, Hobbes' fortunes were secure. But the restored monarch still faced a divided country sapped of material and moral energies, the very condition that had produced England's long season of despair and that had encouraged Hobbes to write *Leviathan*. Thus, as with Francis Brown earlier and John Locke later, Hobbes is found attempting to oppose the voices of doom, the prophets of apocalypse:

> The *Present* onely has a being in Nature; things *Past* have a being in the Memory onely, but things *to come* have no being at all.[23]

His goal is to establish the principles by which, in a word, civil strife might be averted. His method requires a determination of those aspects of human nature that conduce to war and peace and those instruments of governance that will work on and with this nature in such a way as to ensure national tranquility. The metaphor of the machine is adopted (although it is to be doubted that Hobbes considered it a metaphor), and the laws of society are to be fathomed in the same way that Kepler and Galileo discovered laws in physics.

It was Hobbes' belief that a science of society could be established with the same rigor and sureness enjoyed by the science of mechanics. Since humans possess reason and thereby can come to know of cause-effect relations, there is no excuse for their continuing to live in the perilous and uncertain way endured by their ancestors. That they do so is simply to demonstrate that even a rational creature is given to absurdities. Hobbes considered the cardinal absurdities to include the following: (1) a lack of *method* such that we attempt to reason before we have agreed on the meaning of the terms we are reasoning about; (2) confusing the immaterial with the material such that we speak of "infusing Faith," not recognizing that "nothing can be *powered* or *breathed* into any thing but body; and that *extension* is *body*"[24], (3) a continuing belief in the reality of universals; (4) the use of metaphor instead of reality; and (5) a devotion to scholastic terms such as *hypostatical, eternal-Now,* etc., which have no basis in experience at all. He will have none of these. "Words may be called metaphoricall; Bodies and Motions cannot."[25] Hobbes will speak of bodies and motions only. In other words, if man is to be understood scientifically, he is to be understood as matter in motion; if society is to be understood scientifically, it is to be understood as men in motion. Hobbes' plan is to establish first a kind of biophysical behaviorism and, from this, to deduce sociology.

The scientific analysis of man—that is, the biophysical behaviorism—begins in Chapter VI of *Leviathan.* The major variable—what might properly be called the seventeenth-century variable—is *motion.* Hobbes examines animal motion and notes that it takes two forms: *involuntary* ("vitall") and *intentional* ("animal"). The latter, which is the cause of all our grief and joy or very nearly so, is the motion "first fancied in our minds" before being executed by our limbs. While we cannot observe directly the source of this motion, it is none other than motion within the interior parts of the body, motion seeking to satisfy needs of the body. Thus, the motion is caused by appetites that can be subsumed under the label *desire.* Some appetites are native, such as those for food and drink and the avoiding of pain. But the vast majority of our desires "proceed from Experience."[26] The most basic desire, of course, is for the preservation of life. We judge ultimate good and ultimate evil in terms of our own

Protects us from death

Good + Evil - survival

desires, and, since the chief of these is the desire for life itself, we will judge as good or as evil that which promises to preserve or that which threatens our lives. To live, we must have access to the necessities of life: nourishment and protection. Since resources are limited, we are, in the absence of government, in a state of war with our neighbors. We seek power in order to avert our own destruction. We value that which possesses power or confers power upon us, and this is true of the value we place on our fellow men. "The *Value* or WORTH of a man, is as of all other things, his Price, that is to say, so much as would be given for the use of his Power."[27] What we call *dignity* is no more than this. The commonwealth prizes a man for what he can do for them. His status, then, is neither absolute nor permanent. It lasts as long as his power does.[28] What is *honorable* is victory; dishonourable, defeat. Worthiness is fitness.[29] Our desire for power motivates us to search for causes, that is, to acquire science. When we lack this, we must content ourselves with the advice of others. It is in our search for ultimate causes that we discover God and, although we can never know the nature of God, we can and do infer His existence as a blind man infers that there *is* fire when told that fire warms us and when he then feels the warmth.[30] To the extent that religion grants power, it is judged as good and is desired. It will vary across cultures and be affected by the particular circumstances being faced by a given people.[31]

It is often surprising to the modern, Western reader who learns of Hobbes' devotion to monarchy and to the absolute powers of the state to discover that Hobbes arrived at this position from the assumption of the natural equality of all humanity. In fact, *Leviathan* is an object lesson to those who assume that a given philosophical or ethical bias logically entails a given political or social program. The argument for natural equality is presented at the beginning of Chapter XIII, and a lengthy quotation here is warranted:

> Nature hath made me so equall, in the faculties of body, and mind; as that though there bee found one man sometimes manifestly stronger in body, or of quicker mind than another; yet when all is reckoned together, the difference between man and man is not so considerable, as that one man can thereupon claim to himselfe any benefit, to which another may not pretend, as well as he. For as the strength of body, the weakest has strength enough to kill the strongest, either by secret machination, or by confederacy with others, that are in the same danger with himselfe.
>
> And as to the faculties of the mind . . . I find yet a greater equality amongst men, than that of strength. For Prudence is but Experience; which equall time, equally bestowes on all men, in those things they equally apply themselves unto. That which may perhaps make such equality incredible, is but a vain conceit of ones owne wisdome,

> which amongst all men think they have in greater degree, than the Vulgar. . . . For they see their own wit at hand, and other mens at a distance. But this proveth rather that men are in that point equall, than unequall. For there is not ordinarily a greater signe of the equall distribution of any thing, than that every man is contented with his share.[32]

Because of this essential equality among men, there is no sufficient barrier to the assaults of one on another. That is, were there glaring inequalities in ability, the stronger would act with impunity; the weaker would submit without opposition. It is only because each has approximately the same chances of success as the other, each individual against each other and each group against another group, that the seeds of war are planted everywhere and for all time. Natural equality creates confidence and enmity. These, however, create the possibility of a violent death, dreaded by all. Thus, the desire for security calls upon each citizen to invest personal power in the authority of a monarch. The duty to the sovereign lasts as long and no longer than "the power lasteth, by which he is able to protect them." [33]

Leviathan, in a truly remarkable way, integrated the major developments in seventeenth-century philosophy on both sides of the English Channel. With Descartes, Hobbes identified matter with extension. Siding with the Gassendists, he insisted that only body can affect body and that only matter in motion can serve as the subject of scientific inquiry. While applauding Bacon's *Novum Organum,* he saw in the works of Galileo what true science was to become: a rational approach to nature in which theory serves always as the handmaiden of fact.

Hobbes, for reasons that must be clear, was greatly respected by the nineteenth-century utilitarians. His program was pragmatic, mechanistic, objective, and, in a subtle way, egalitarian. He could find no source of human conduct other than the desire to survive and prosper. We cannot be sure that the Epicurean revival launched by Pierre Gassendi was seminal in this respect, but the presumptive evidence is substantial. His psychology was materialistic, hedonistic, behavioristic. The mind, while not doubted, is subject to the laws of physics just as is the rest of the body! To know the causes of one's actions is to know one's desires, to know one's needs. Prime among these are the basic, biological requirements for survival. All other needs, acquired by experience and tradition, finally derive their force from these. In the introduction to *Leviathan* he wrote, "*Reward* and *Punishment* (by which fastned to the seate of the Soveraignty, every joynt and member is moved to performe his duty) are the *Nerves,* that do the same in the Body Naturall . . . *Concord, Health; Sedition, Sicknesse;* and *Civill war, Death.*" [34] Plato had examined the Republic in order to enlarge the canvas enough so that the nature of man could be discerned. Hobbes, too, saw the state as but an "artificial man." The difference, of course, is that Plato judged the ends of the state in terms of virtue; Hobbes, in terms of utility. Plato's metaphor was spirit; Hobbes', the machine.

Hobbes may be said not only to have initiated one of the first and most influential systems of materialistic philosophy in modern times but also one of the least compromised versions of *egoistic* ethics. His emphasis upon the motive for personal survival reappears continually as he addresses questions of morality. Hobbes is persuaded that what passes for benevolence, altruism, and other expressions of a principled regard for others must finally be analyzable in the terms of private gain and self-regard.

In Chapter 2, we paused to examine a classical defense of *egoism*, the one presented by Thrasymachus as an alternative to Socrates' ultrarational theory of benevolence. Thrasymachus can find no reason to believe that the holder of Gyges' ring will decline to avail himself of that which invisibility allows. Why would anyone defer the gratification of each and every desire except out of fear of reprisal? What reason has Smith to come to the rescue of Jones other than the possibility of reward or because he seeks to establish grounds for reciprocation?

The Socratic relaxation of the apparent tension between self-regard and altruism is based upon the theory that an enlightened individual can find happiness only in those actions that are good in themselves. Therefore, actions devoid of benevolence or indifferent to the needs of the state cannot produce happiness in the enlightened. Note, however, that this attempted solution plays readily into Hobbesian hands. Hobbes, even if he accepted the Socratic theory, is still able to insist that the happiness of the individual survives as the root-motive. Smith may, indeed, perform an act in the interest of the state but he does so because it brings him pleasure.

materialistic philosophy

As with Hobbes' materialistic philosophy, his *egoistic* ethics would also undergo revision. Forms of it persist in all utilitarian theories and in those theories of value usually classified as "situation ethics." The eighteenth-century sentimentalist or "empathy" theorists—Bernard Mandeville and Francis Hutcheson have already been cited—could not eschew the appeal or the challenge of egoism. The individual might well be so framed as to possess an inborn concern for others, but what is this concern other than the pleasure produced by actions of a certain kind? And from the proposition that value-laden thought and conduct are inescapably selfish, it is a small step to the proposition that one must tailor for oneself that set of tastes, opinions, and dispositions conformable with one's own character, one's *ego*. Here, of course, are the seeds of individualism and liberalism. From these would grow the overarching aims of personal growth and personal success that define the commercial tone of Victorian England. It was these elements of Hobbes' ethics that would reappear in Tom Paine's leaflets, in the antiroyalist literature of revolutionary France, in the British Reform Act (1832), and nearly every day in our own time as the liberated citizen celebrates the multiplication of rights and the lessening of duties. If one

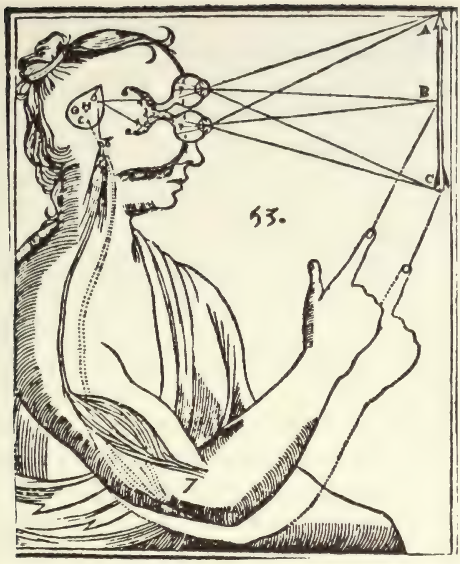
If some beh's are automatic, why not all? How can we tell what's intentional? (when, how, + if does will come in?)

were to judge from the persisting and even growing signs of disillusionment displayed by those who have received the greatest share of egoistic benefits, one might conclude that Socrates had sensed something of a deep and durable nature, something missed or misunderstood even by Hobbes. Can a human being find pleasure in acts which themselves are not good?

The Reflex

Ours is a time of scientific specialization and, as the term suggests, "professionalism." When we read a history of psychology, we expect to find the names of Freud and Wundt and the usual philosophers. We are ready to admire a Descartes who, with all his concern for philosophy, still had time to contribute to optics and geometry; a Locke whose interests in government did not prevent him from providing a theory of the mind; a Galileo whose imagination was able to embrace astronomy, mechanics, and the philosophy of science. Our admiration takes the form of orphaning these figures as "universal geniuses," exceptions to the specialist's rule. While this attitude is understandable, it is not correct historically. If Locke has been an influential figure in the history of psychology, and he certainly has, then so also was Newton. The scholars of the seventeenth century were not "professionals." In the important respects, they assumed the unity of science to an extent the modern epoch hardly approximates. Newton, for example, was driven to a theory of vision and, particularly, to a theory of color vision, because he believed that the universal law of gravitation was *universal* and that its effects should be as apparent in a biological system as in planetary motion. Borelli, a colleague of Galileo's, studied the physiology of muscles in order to apply Galileo's principles of mechanics to systems that just happened to be alive. Descartes and Hobbes both sought to base their epistemologies upon the new science that was mechanics. Descartes invented the physiological theory of reflexes with this in mind. To him and to his scientific contemporaries, it would have been inconceivable that there could be a "behavioral science" that was not, by the very fact, physics. Even Locke and Hume, disinclined to speculate on the biological basis of mind (disinclined, we may assume, because of the heavy weather suffered by Descartes at the hands of the Gassendists), both agreed that the issue was to be settled by "the anatomists." That is, while not addressing the question of materialism directly, Locke and Hume recognized it to be a *scientific* question. Hume was explicit:

> [W]hen the circulation of the blood . . . is clearly proved to have place in one creature . . . it forms a strong presumption, that the same principle has place in all. . . . [A]ny theory . . . of the understanding, or the origin and connection of the passions in man will acquire additional authority, if we find, that the same theory is required to explain the same phenomena in all other animals.[35]



Descartes' Optical Model of Perception

Descartes had said very much the same but had excluded animals from the domain of reason. Gassendi, however, had challenged this restriction, arguing that animals display memory, desire, and even a certain inductive capacity. Given the scientific climate of the second half of the seventeenth century, Descartes' dualistic path seemed to many to be too winding. When Newton's physics triumphed over Descartes', the latter's authority in all subjects fell into question, if not disrepute. His theory of the reflex had made action the consequence of stimulation and was designed to embrace a great part of all our conduct. Newton's laws, able to describe the motion of everything else, were judged by some to be immediately applicable to animal motion. (Recall Voltaire's remark, and it was Voltaire who brought Newton's work to the attention of the French.)

Descartes had envisaged the mechanism of the reflex to operate in the manner illustrated in the above figure. The arrow's image is conveyed by the optic nerves to the brain. The inverted image is set aright within the optic chiasm whence it passes to the pineal gland. Motor nerves receive "spirits" from the pineal gland in amounts determined by the force of the visual impression. Descartes considered the mechanism by which the nerves are actually energized by the spirits to be unfathomable.

The *Treatise of Man* was a work that Descartes would not allow to be published in his lifetime. The work first appeared in 1662, twelve years after his death and seven years after Gassendi's. It is his most biological work, and it is in this treatise that the lurking monist in Descartes comes close to surfacing. The *Treatise* invokes the idea of a model or "machineman" with a body like a statue. To this statue, anticipating Condillac, Descartes adds powers and functions by equipping it with sensory, motor, and reflex mechanisms. The essay includes the operation of feedback, emphasizes the muscular foundations of attention, and relates the quality of our experiences (or, the machine's experiences) to the action of the spirits flowing in and out of the pores of the brain. The spirits are rendered particulate, varying in mass and other qualities. As a result of these material differences, the spirits are able to mediate different emotions and the actions deriving from these emotions. Except for the faculty of reason, this machine simulates nearly perfectly a real man:

> I desire you to consider, I say, that these functions imitate those of a real man as perfectly as possible and that they follow naturally in this machine entirely from the disposition of the organs—no more nor less than do the movements of a clock. . . . Wherefor it is not necessary, on their account, to conceive of any vegitative or sensitive soul or any other principle of movement and life than its blood and its spirits, agitated by . . . those fires that occur in inanimate bodies.[36]

Newton's laws of motion were ideally suited to account for such a system:

1. Every body continues in its state of rest or of uniform motion in a right line unless it is compelled to change that state by forces impressed upon it.
2. The change of motion is proportional to the motive force impressed and is made in the direction of the right line in which that force is impressed.
3. To every action there is always opposed an equal reaction; or, the mutual actions of two bodies upon each other are always equal and directed to contrary parts.

Today we accept these laws as determining the manner in which objects will react to imposed forces. In the seventeenth century, these were laws of "science," potentially as applicable to the mechanics of mental life as to billiard balls. Descartes' notion of "animal spirits" was transformed into the *vis nervosa* (nervous *force*), and the laws of the reflex were relegated to the science of physics. Increasingly, English empiricism and French materialism looked alike. The Gassendists and other anti-Cartesians could assert the failures of Cartesian physics (while subscribing to his theory of reflexes), and the empirical "experimental philosophers" in England could begin the grand program of reducing philosophy to science. In both quarters, the concept of the reflex was central. British empiricism had been, since Locke's *Essay,* associationistic. By

Made many discoveries in biology
of human body + mind.

the eighteenth century it was more compelling to some to speak not of the association of ideas but of a more mechanical association in a more mechanical entity. Here and there, experiments began to address the physiology of this reflex and, hand in hand, these experiments and entire mechanistic theories of psychology marched toward the century of materialism, the nineteenth.

Perhaps the most significant discovery of the eighteenth century in regard to the evolution of psychological materialism was made by Luigi Galvani, who in 1786 reported the results of experiments on the stimulation of the muscles of frogs by the application of an electrical pulse. Until the essentially *electrical* nature of nerve-muscle interactions was discovered, the mystery in Descartes' system was preserved. David Hartley (1705–1757) had already published his *Observations on Man*,[37] which had presented psychology as the science concerned with the mechanics of associationism. Hartley's thesis was self-consciously Newtonian, and prematurely so. Thomas Reid was only one of several philosophers to dismiss Hartley's "science" as having no basis whatever in observable fact. Thus, only with Galvani's discovery was it possible to advance a mechanistic psychology equipped with an actual mechanism. Such a psychology needed the biological equivalent of the colliding billiard balls, something tangible and measurable. Galvani did not discover the neural impulse itself, but he did establish that muscles could be caused to contract by the application of an electrical charge. The effect was dismissed by a number of scientists—including the great Alessandro Volta—who insisted that the human body was not even able to conduct a charge. This complaint was ingeniously set aside by one of the more elegant demonstrations in the history of the neural sciences. A small boy was suspended by a long rope. Hung from a separate beam were leaves of gold, placed just in front of the boy's nose. Then, a glass rod was rubbed briskly with cat's fur and placed against the boy's bare toes. The gold leaves were immediately drawn to the boy's nose, and the theory of electrical conduction by the human body entered the realm of indisputable fact. Experiment again had triumphed over dogma.

At about the same time that David Hartley was insisting with Hobbes, but without evidence, that "each action results from the previous circumstances of the body and mind, in the same manner and with the same certainty as other effects do from their mechanical causes,"[38] Robert Whytt (1714–1766) was repeating an experiment that had been performed years earlier by the physiologist Stephen Hales. Hales had observed that the decapitated frog could be induced to move its limbs by pinching them. Even without a brain, the frog would avoid a painful stimulus delivered to the extremities. A surgical severing of the spinal cord, however, eliminated the response. In his essay "On the Vital and Other Involuntary Motions of Animals" (1751), Whytt argued that the spinal cord was able to mediate stimuli and responses in the absence of activity of the brain.[39] What Hartley was speculating, Whytt was demonstrating and in so compelling a way that a Newtonian biology seemed to be merely a matter of time.

The concept of "association," and its mechanical equivalent, the "reflex arc," had the same status in eighteenth- and nineteenth-century physiology as "attraction" had in the physics of the seventeenth and eighteenth. Hartley illustrates the inextricable tie joining the emerging psychology to the established physics in the opening chapter of *Observations on Man:*

> My chief design . . . is briefly to explain, establish, and apply the doctrines of *vibrations* and *association.* The first of these doctrines is taken from the hints concerning the performance of sensation and motion, which Sir Isaac Newton has given at the end of his Principia . . . the last from what Mr. Locke and other ingenious persons since his time have delivered concerning the influence of *association* over our opinions and affections.[40]

The essay then goes on to present ninety-one propositions regarding mental life, propositions embracing the simplest sensations and others designed to explain language, emotion, and dreams. All are based on the notion of a mechanical interaction between the organs of sense and the motor systems. Integrating these two systems is the brain. When we are stimulated, vibrations are established in the nerves and conveyed to the brain. After the stimulus has been removed, the vibrations persist, diminishing as a function of time. Memory, by this view, is the fading vibration induced by a bygone event. Learning is the establishment of a connection of what Newton had called "subtle forces" and Hume had described as the "gentle force": that is, a connection between *physical* events in the brain. The following theorem very nearly says it all:

> If any sensation A, idea B, or muscular motion C, be associated for a sufficient number of times with any other sensation D, idea E, or muscular motion F, it will, at last, excite d, the simple idea belonging to the sensation D, the very idea E, or the very muscular motion F.[41]

Hume had said as much. Events occurring together, in unaltered sequence, often will come to be treated as causally related. Descartes, in describing the mechanism of nonspiritual action, had already specified the anatomical arrangements by which such sensory-motor events take place. Newton's "vibratiuncles" were but the earliest form of a *vis nervosa.* Hartley even finds support in Leibniz's "pre-established harmonies," which free the philosopher from having to explain how an immaterial soul directly activates a material body. Hartley's essay is, then, one of the great syntheses of the eighteenth century. He combines the physics (and method) of Newton with the physiology of Descartes; he avoids theological dilemmas by invoking the parallelism of Leibniz; he takes for granted the empirical associationism of Locke and Hume; he assumes, as Hume and every cobbler in England did, that pleasure and pain are the glue that binds our associations. Perhaps more than any single work previously published, Hartley's *Observations on Man* is a treatise in *modern* psychology. Its influence in the nineteenth century, as we shall see in the next chapter, was considerable.

"L'Homme Machine"

The luminaries of the French Enlightenment, whether atheists or theists, whether monarchists or anarchists, all shared in their contempt for dogma. Voltaire was the center and the engine of the movement. He admired England, wrote glowingly of the English people, praised them for their egalitarianism, their opposition to the Roman Church, and their tolerance. While history reveals that Voltaire's plaudits were overly generous, there is no doubt but that the Catholic influence in eighteenth-century France was oppressive. The Enlightenment in France, which cannot be said to have produced philosophic perspectives of lasting significance, was essentially a *political* movement posing as an intellectual enterprise. It is not surprising that the most original philosophical thinker in the group was Rousseau, and that his only important contribution to philosophical thought is to be found in his additions to the "social contract" theory.

As political figures, the men and women of the Enlightenment saw in the new science and philosophy of their age a weapon that could protect them against the excesses of orthodoxy and, more, could drive a wedge between the received truths of religion and the complacent mass of believers. Politicized philosophy, like "political art," may have sudden effects, but not enduring ones. It may move people, but it does not advance the discipline in which it pretends membership. Voltaire himself introduced Newton to the French intellectuals but never fully comprehended Newtonian physics. Much of his competence in this subject was the gift of Mme. du Chatelet. This is not to say that Voltaire merely translated the *Principia* and chanted its laws. Rather, it is only to note that Voltaire's real interests in Newton's physics were extrascientific. He found in Newton another reason to oppose Cartesianism. The same may be said of Condillac (1715–1780), who translated Locke's *Essay* into French and whose *Essay on the Origin of Human Knowledge* remains one of the best analyses of Locke's empirical philosophy. It was Condillac who proposed the "sentient statue," standing in the path of a universe of stimuli, taking on psychological characteristics as a result of experience. It was Condillac's "statue" that served as the template for much of the materialistic philosophy of late eighteenth- and early nineteenth-century France. But Condillac was not particularly interested in the material basis of mind, and less in physiology. Nor was he simply a disciple of Locke's.[42] Rather, he was captivated by the anti-

Dogma – a body of theological doctrines strictly to be attached to

metaphysical alternative of the English school, and by the philosophy of experience celebrated by Voltaire and no less so by Condillac's own cousin, d'Alembert. Again, the pull was not simply toward England but away from Descartes. Condillac was a priest. His enemy was clearly not the Church. His enemy was dogma, and especially philosophical dogma disguised as religion.

We have seen that Aristotelianism was an early casualty of the French religious wars. Galileo's "new sciences" were judged by the men and women of the Enlightenment as the definitive refutations of Peripatetic philosophy, although it is to be noted that their understanding of Galileo's physics was fuller than their understanding of Aristotle's *Metaphysics.* Cartesianism soon loomed as authoritarian as the *ism* so recently rejected. French intellectuals were now unanimous in their opposition to any assertion that flew in the face of fact and to any proposition not supported by the data of experience, excepting only those propositions set forth softly and humbly in the language of simple faith. No longer would they accept Thomism or Cartesianism as scientific. More significantly, they would no longer base government, social discourse, or the everyday affairs of life on foundations whose only claim to attention was custom or authority.

It is frivolous to date an idea and reckless to date large social movements. The eighteenth century was a century of revolution, but so was the sixteenth and, indeed, the seventeenth. The social, political, and economic changes that began most visibly in that period we so easily isolate as the Reformation are changes still taking place in our own time and often with revolutionary zeal and revolutionary consequences. Thus, it can only be apologetically that one advances the thesis that Gassendi, Newton, Voltaire, Rousseau, and their lesser contemporaries created a revolution in perspective that culminated in a revolution of classes. It is better and more accurate to put the thesis this way: The Reformation was an attack on institutions and this attack came to embrace the very idea of authority. With Hobbes, there was introduced the summoning utilitarian notion that the justification of governments is to be found in the safety they confer on human life. Galileo and Descartes, as scientists, contributed to the demise of that Aristotelian bastion which, for so long, had protected the claims of institutional authority. Descartes' additional contribution was that of publishing his major works in vulgar French, a policy embraced by every major French philosopher to follow. Locke, Hume, and their Continental counterparts, Gassendi and Condillac, translated the social reality into enduring philosophical systems. This is not to say that they were apologists for a movement. It is, however, to avoid the suggestion that they were the causes of the movement. They were, we prefer to say, participants and very influential ones at that.

We noted at the beginning of Chapter 7 that empirical philosophy does not *entail,* in the logical sense, materialism. (Berkeley was an ungrudging empiricist and a devout immaterialist.) But for the person who believes that the mind is furnished by experience alone, for the philosopher who can find no source of knowledge beyond the senses, the step to psychological materialism is short, logic to the contrary notwithstanding. Condillac's metaphor of the statue is a *material* metaphor. Hartley's vibrations—never witnessed by Hartley or anyone else—were not even presented as metaphor but as unseen reality. Recall that Locke and Hume specifically eschewed the temptation to theorize on the material causes of sensation and association. Hartley, a contemporary of Hume's, already displayed the attraction materialism always holds for the empiricist. Hobbes succumbed to it on the authority of Galileo; Hartley, on the authority of Newton's physics. In fact, there is a sense in which Descartes' dualistic psychology is understandable in terms of its not being an extension of an empirical philosophy.

In 1748 a book appeared bearing the title *L'Homme Machine*.[43] Its author, Julien Offray de La Mettrie (1709–1751), was a physician, a self-exiled member of the court of Frederick the Great, and the boldest psychological materialist France had produced. The book caused a commotion even greater than that produced by his *Historie Naturelle de l'Âme* (1745), whose sudden infamy had caused La Mettrie to move to Bavaria. Both books were relatively short, highly polemical, and hardly either philosophical or scientific as these adjectives are properly assigned. It is in *L'Homme Machine* that the soul is reduced to "an enlightened machine" and the human psychological faculties reduced to mere brain physiology. La Mettrie's "proofs" for these assertions are drawn from such otherwise irrelevant observations as the ability of a chicken to continue to run after its head has been removed, and the fact that the freshly removed heart of an animal will "jump" out of boiling water. His authorities include military officers who have told him tales of battle casualties but also include several of his own clinical observations. There is nothing in either book that the modern neurophysiologist or physiological psychologist would take very seriously, and less what modern philosophers with their wits about them would pause to reread. The importance of *L'Homme Machine* is historical but not because it was especially influential, beyond the influence enjoyed by any book officially condemned. It is important in its style: porous, haughty, devoid of self-critical strains. It is the sort of book one writes on a dare or as a challenge. It is defiant but seldom clear on the issue or opinion to be defied. If the enemy were those who insisted that man is endowed with an immortal and incorporeal soul, the fact that chickens run after decapitation is utterly irrelevant. If the antagonist was the Cartesian who believed that the will is directed by the soul and that the soul directs its influence from a region of the brain, the behavior of hearts dropped into water is beside the point. La Mettrie, in summarizing the respects in which man is matter, went on to insist that he is only matter, but this, as we know, cannot be established by materialist methods without committing the fallacy of *petitio principii*.

If La Mettrie had a direct influence on his successors it was on Pierre Cabanis (1757–1808), one of Napoleon's senators and a leading light in the French materialist movement. Both La Mettrie and Cabanis, however, have received more attention in histories of psychology than history warrants. It was Gassendi and the Gassendists who founded psychological materialism. Even Hobbes cannot compare to them in immediate influence. La Mettrie and Cabanis, both physicians and both lacking in that subtlety of mind that philosophy demands, contributed little to the historic debate between materialists and dualists, and nothing to the science spawned by the former. Both were products of the Gassendist tradition and both displayed the neo-Epicurean attitude of Gassendist philosophy. For both, happiness and morality, no less than sensation and action, were to be comprehended in material terms and were subjects of scientific analysis. Neither authored a bona fide materialist system. Indeed, neither had Gassendi. The system, of course, was there all along. It was the system of Galileo and Newton. But unlike Galileo and Newton, the French materialists were not physicists, were not incisive philosophers and, by and large, were not even scientists. Uncritically, they assumed psychophysical isomorphism and were thereby convinced that the laws of physics, the laws of society, and the laws of psychology were but three versions of the same material principles. Ironically these writers—d'Alembert, La Mettrie, Cabanis—are closer to a dominant contemporary perspective than were any of their more illustrious brothers. What they sensed, and what either eluded or embarrassed or failed to intrigue Locke, Hume, Leibniz, and Kant, was the growing possibility of wedding philosophy to anatomy. They were an eager group, forcing compatibilities where there was no ground other than enthusiasm for even speculating on the compatibilities. They saw into the next century.

The Scientific Context

Psychological materialism did not appear in a vacuum, nor is it to be understood simply as a consequence of the Gassendist rebuke of Cartesianism. We have devoted most of our attention to Galileo and Newton, by far the most important scientists of the seventeenth century and the men who gave the greatest impetus to eighteenth-century science. But there were many other, though smaller, contributions to the scientific spirit. La Mettrie, for example, studied with the great Dutch physician Hermann Boerhaave (1668–1738), whose treatises in chemistry had rejected vitalism and had insisted that all life processes were reducible to chemical processes. Antoine Lavoisier (1743–1794) convincingly demonstrated the conservation of matter in an experiment in which the steam from boiled water was condensed and collected. The notion of matter as being "neither created nor destroyed" is a late eighteenth-century one (discounting, as usual, the ancient Greek writers) and one with obvious religious implications. The same Lavoisier advanced the theory of chemical elements, and, in a philosophical climate, this theory held out the promise of reducing the mysterious diversity of nature to its building blocks. We might underscore the growing materialism and scientism of La Mettrie's era by noting, without discussing, the scientists who were active and prominent at the time: Joseph Black (1728–1799), Charles Coulomb (1736–1806), Benjamin Franklin (1706–1790), Luigi Galvani (already cited), Stephen Hales (1677–1761), Albrecht von Haller (1708–1777), Karl Linnaeus (1707–1778), and Joseph Priestley (1733–1804). It was the scientific achievement of this group that convinced some of the finest minds of the nineteenth century that the abiding questions could be answered at last and answered positively.

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11. Ibid., p. 339.
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13. Ibid., p. 443.
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28. Ibid.

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32. Ibid., Pt. I, Ch. 13.

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